

**University of Colorado at Colorado Springs
Department of Mechanical and Aerospace Engineering**

SPCE 5065

HOMEWORK #3

DUE: Friday 10 July 2026

SOLUTION BY L. GEORGE

Instructions:

- Include your references at the end of your homework in APA format.
<https://www.aiaa.org/publications/journals/reference-style-and-format>
- Include units in your final answers!
- Write non-mathematical answers in complete sentences.
- Clearly state your assumptions

1. For the current events presentations on Thursday 2 July

- a) Summarize the presentation
- b) Describe something you learned from it
- c) Write one question you have left about the presentation

2. What is meant by “altered vestibular functions” and what are its symptoms?

Altered vestibular functions refers to changes in the body's balance and spatial orientation system caused by the absence of gravity. On Earth, the vestibular organs in the inner ear use gravity to help determine which way is up and to coordinate balance, posture, and eye movements. In microgravity, these organs receive conflicting sensory information, leading to symptoms collectively known as Space Adaptation Syndrome. Common symptoms include dizziness, nausea, motion sickness, disorientation, vertigo, headaches, impaired balance, and difficulty coordinating movements. Most astronauts adapt within a few days, but these effects can temporarily reduce performance during critical mission tasks.

3. What are approximate caloric intake requirements for astronautics? Describe two unique nutritional requirements driven by the free-fall environment.

Women:

$$BEE = 655 + (9.6 \times W) + (1.7 \times H) - (4.7 \times A)$$

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Men:

$$BEE = 660 + (13.7 \times W) + (5 \times H) - (6.8 \times A)$$

W = weight in kg

H = height in centimeters

A = age in years

Their source should be referenced.

4. For a Mars mission, what habitable volume would you recommend for the crew quarters on the

- a. Flight there
- b. Surface
- c. Return flight

Explain your rationale.

a. 20 m³ – recommended by NASA for 6-month flight

b. 100 m² – Necessary for human mental health, exercise areas, personal areas, airlock, and lab areas for analysis of Mars samples

c. 25 m³ – more for continued analysis of Mars samples on the return flight

Answers may vary provided good rationale is included.

5. NASA's Bioastronautics Roadmap provides overall ratings for a human Mars mission as shown in the table below. Details may be found here [Human Research Roadmap \(nasa.gov\)](https://www.nasa.gov/human-research-roadmap).

- a. Complete the columns for a Ceres mission.
- b. Choose three specific areas and discuss them more thoroughly. How would the risk be different for Ceres? What might be a mitigation strategy for that risk?

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Risk	Mars operational	Mars long term	Ceres operational	Ceres long term
Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders				
Risk of Adverse Health and Performance Effects of Celestial Dust Exposure				
Risk of Adverse Health Effects Due to Host-Microorganism Interactions				
Risk of Adverse Health Event Due to Altered Immune Response				
Risk of Adverse Health Outcomes and Decrements in Performance Due to Medical Conditions that occur in Mission, as well as Long Term Health Outcomes Due to Mission Exposures				
Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		N/A		
Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks				
Risk of Bone Fracture due to Spaceflight-induced Changes to Bone				
Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes				
Risk of Impaired Performance Due to Reduced Muscle Size, Strength and Endurance				
Risk of Ineffective or Toxic Medications During Long-Duration Exploration Spaceflight				
Risk of Injury and Compromised Performance Due to EVA Operations				
Risk of Injury from Dynamic Loads				
Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		N/A		
Risk of Performance Decrement and Crew				

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Illness Due to Inadequate Food and Nutrition				
Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload				
Risk of Radiation Carcinogenesis				
Risk of Reduced Physical Performance Capabilities Due to Reduced Aerobic Capacity				
Risk of Renal Stone Formation				
Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)				

Answers will vary, but should be clearly communications and properly referenced.

6. Describe the major breakdowns and disconnects that occurred during the Apollo 13 mission. Which elements of the SHELL model were involved?

Answers will vary, but here is one example:

The Apollo 13 accident resulted from a combination of technical failures and human-system disconnects rather than a single mistake. The immediate cause was an oxygen tank explosion caused by damaged wiring inside the tank, but underlying causes included design flaws, inadequate testing after modifications, and communication breakdowns between engineering teams. Once the explosion occurred, the crew lost electrical power, oxygen, and much of the command module's functionality, forcing them to rely on the lunar module as a lifeboat. Throughout the emergency, successful recovery depended on effective communication and coordination between the astronauts, Mission Control, engineers, and contractors.

Within the SHELL model, several interfaces were involved:

- Liveware–Hardware (L-H): The oxygen tank design, damaged wiring, thermostat incompatibility, and failure of critical spacecraft hardware were the primary technical failures.
- Liveware–Software (L-S): Testing procedures, operational documentation, engineering analyses, and mission procedures failed to identify the hazards associated with the modified oxygen tank.

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- Liveware–Environment (L-E): The space environment left the crew with no opportunity for repair or rescue, while limited power, oxygen, and thermal control greatly increased mission risk.
- Liveware–Liveware (L-L): Effective communication among the crew, Mission Control, and engineering teams became the key factor that enabled the safe return of the astronauts despite the equipment failures.

Apollo 13 demonstrates that successful human spaceflight depends not only on reliable hardware, but also on well-designed procedures, effective communication, and the ability of humans to adapt when unexpected failures occur.

7. Propose requirements for an EVA suit for a human mission to the following destinations. Include a discussion of the unique aspects of the destination.

- a) Moon
- b) Mars
- c) Ceres

Answers will vary but should be well described and referenced properly.

8. NASA has allocated an additional 1,500 kg of launch mass for improving crew health during a 900-day Mars mission. You may invest the mass in only three of the following:

- radiation shielding
- exercise equipment
- food variety
- medical equipment
- private crew quarters
- water reserves
- artificial gravity demonstration hardware
- additional scientific equipment.

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Develop a weighted trade study and justify your allocation. Make sure your answer includes:

- objectives
- evaluation criteria
- weighting factors
- trade matrix
- final recommendation

Answers will vary but should be clear and properly referenced.

9. Using the SHELL model, compare two human-spaceflight accidents. Evaluate as applicable:

- common failure modes (in the context of the SHELL model)
- organizational influences
- design deficiencies
- operational decisions
- lessons that remain relevant for Artemis or Mars

Answers will vary depending on missions chosen, but should be clearly written and properly referenced.